



# Approximation Algorithms for Big Data

## Lecture 10: Low Rank Approximation

Spring 2026

Instructor: Tal Wagner

# Administrative Announcements

- University-wide teaching survey is live:
- [https://my.tau.ac.il/TAU\\_Student/TeachingSurvey](https://my.tau.ac.il/TAU_Student/TeachingSurvey)
- Your feedback is highly valuable

הדקנט לחדשנות  
בהוראה ובלמידה  
אוניברסיטת תל אביב



סקר שביעות רצון מהוראה  
סמסטר ב', תשפ"ז

**יפתח מחר, 23/6 בשעה 8:00 למענה**

ספרו לנו, איך עבר הסמסטר?



מקנה  
נקודות  
בבידינג

**הקליקו לסקר**

כניסה למידע האישי > מענה על הסקר

הסקר יימשך שלושה וחצי שבועות  
ויהיה פתוח למענה עד ל-18/7/26

# Administrative Announcements

- **Next week** (July 9): Guest lecture by **Dori Eldar** from **Mobileye**
- **Topic:** probabilistic framework for aligning multiple vehicle drives over a local map
- Details published on email and class forum on Moodle



The poster is for a "SPECIAL GUEST LECTURE" titled "AAA: Alignment, Assignment & Aggregation" with the subtitle "A Principled Approach to Multi-Drive Map Alignment". The speaker is Dori Eldar, Principal Engineer at Mobileye. The event is on July 9, 2026, at 15:00, in the Goldschlager Multidisciplinary Engineering Building, 3rd Floor, Room 315. Logos for Mobileye and Tel Aviv University are at the top.

mobileye™

אוניברסיטת תל אביב  
TEL AVIV UNIVERSITY

SPECIAL GUEST LECTURE

AAA: Alignment, Assignment  
& Aggregation

A Principled Approach to Multi-Drive Map Alignment

 Dori Eldar  
Principal Engineer, Mobileye

July 9, 2026 | 15:00  
Goldschlager Multidisciplinary Engineering Building  
3rd Floor, Room 315

# Administrative Announcements

- Class engagement reporting open on Moodle
  - Please fill by **July 17**

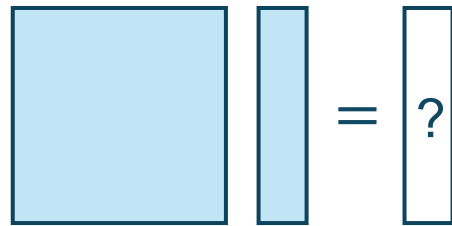
# Plan

- Recap last time (with corrections)
- Today's topic: Approximate low-rank approximation via sketching
- Brief interlude: low rank adaption (LoRA) in LLMs

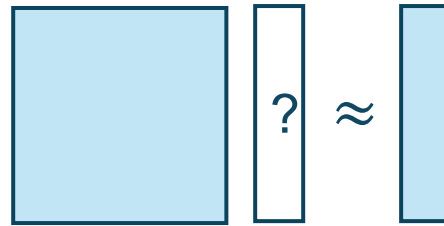
# Recap

# Computational Linear Algebra

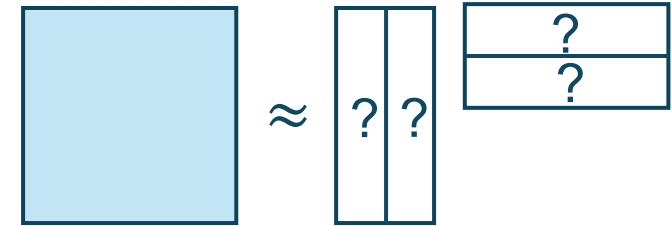
- Extremely useful and prevalent
- Extremely computationally taxing
- We will develop efficient randomized approximation algorithms in the spirit of class
- Why would the techniques we've seen so far be related?



Matrix-vector /  
matrix-matrix  
multiplication



Linear regression



Low-rank  
approximation

# AMM: Upshot

- We proved:
  - $S$  with iid entries from  $\frac{1}{\sqrt{m}} N(0,1)$  with  $m = 1/\varepsilon^2 \delta$  satisfied 2-JLMP:

$$\sqrt{\mathbb{E}[(\|Sx\|_2^2 - 1)^2]} \leq \varepsilon\sqrt{\delta}$$

- 2-JLMP implies AMM:

$$\Pr[\|(SA)^T(SB) - A^T B\|_F \leq \varepsilon\|A\|_F\|B\|_F] \geq 1 - \delta$$

- Therefore:  $S$  with iid entries from  $\frac{1}{\sqrt{m}} N(0,1)$  with  $m = 1/\varepsilon^2 \delta$  satisfies AMM
- Remark: Can get AMM with only  $m = \log\left(\frac{1}{\delta}\right)/\varepsilon^2$ 
  - Can be proved via higher JL moments



# $\ell_2$ -Regression

- Definition: in the  $\ell_2$ -regression problem,
  - Input:  $A \in \mathbb{R}^{n \times d}$  and  $b \in \mathbb{R}^d$
  - Output:  $x^* = \min_{x \in \mathbb{R}^d} \|Ax - b\|_2$  (i.e., solve the linear equation as best as possible)
  - Exact solution?  $x^* = (A^T A)^+ A^T b$
  - Time of exact solution?

- Approximate  $\ell_2$ -regression : Given  $\varepsilon, \delta$ , find  $x$  such that

$$\Pr[\|Ax - b\|_2 \leq (1 + \varepsilon) \min_{x \in \mathbb{R}^d} \|Ax - b\|_2] \geq 1 - \delta$$

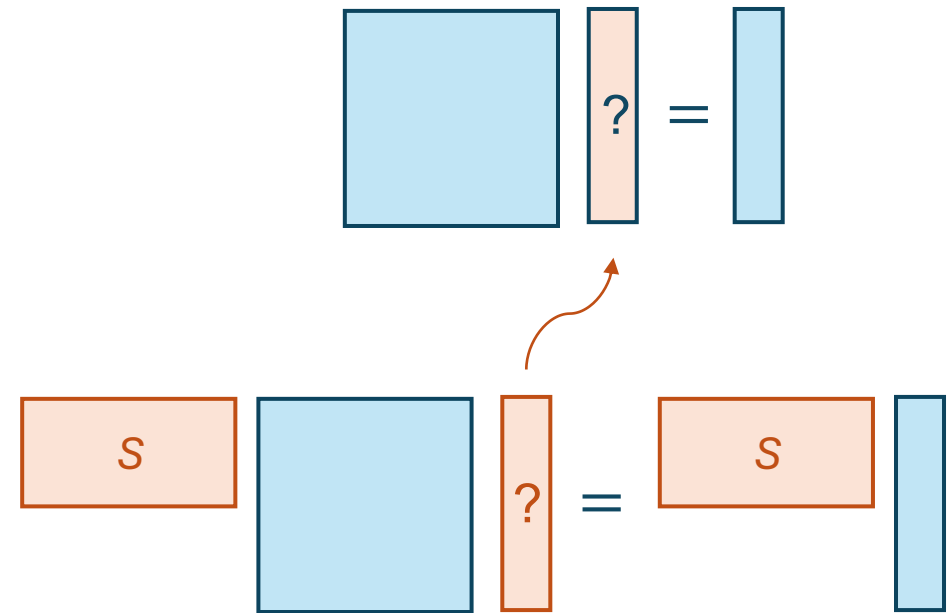
- Ideas how to achieve this?

# $\ell_2$ -Regression via Sketch-and-Solve

- How can we solve  $\ell_2$ -regression with an OSE?

- Sketch-and-solve:

- Sample  $S$
- Solve  $\tilde{x} = \min_{x \in \mathbb{R}^d} \|SAx - Sb\|_2$  exactly
- Return  $\tilde{x}$



# Oblivious Subspace Embedding (OSE)

- **Definition:** a distribution over random matrices  $S \in \mathbb{R}^{m \times n}$  is an  $(\varepsilon, \delta, d)$ -oblivious subspace embedding (OSE) if for  $\mathbb{R}^n$  if for every given subspace  $\mathcal{V} \subset \mathbb{R}^n$  of dimension  $d$ , we have:

$$\Pr[\forall x \in \mathcal{V}, \quad \|Sx\|_2^2 = (1 \pm \varepsilon)\|x\|_2^2] \geq 1 - \delta$$

- *"JL for an entire subspace"*
- **Equivalent definition:**  $S$  is an OSE if for every given matrix  $A \in \mathbb{R}^{n \times d}$  we have:

$$\Pr[\forall x \in \mathbb{R}^n, \quad \|SAx\|_2^2 = (1 \pm \varepsilon)\|Ax\|_2^2] \geq 1 - \delta$$

- Why are these the same?

# OSE $\Rightarrow$ Regression

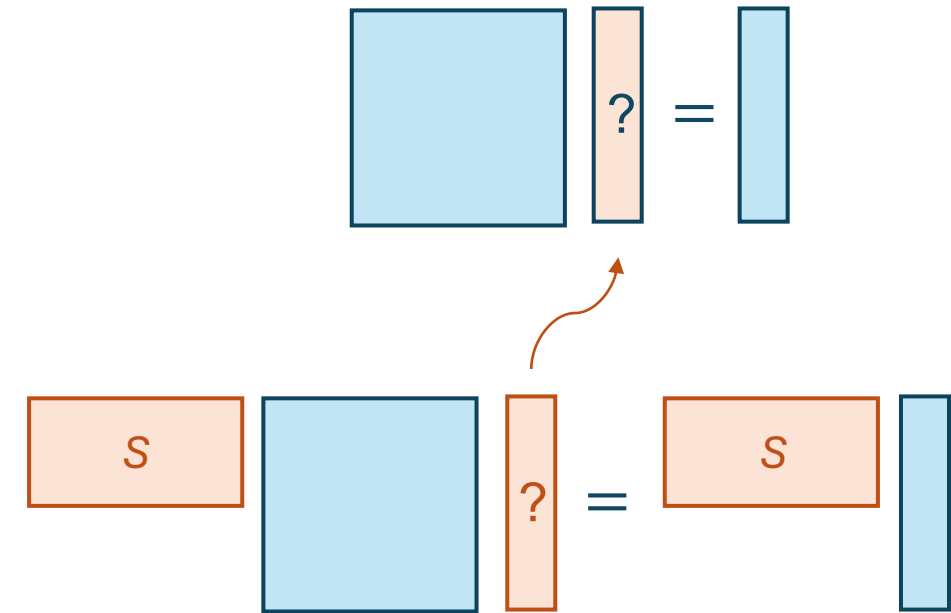
- How can we solve  $\ell_2$ -regression with an OSE?

- Sketch-and-solve:

- Sample  $S$
- Solve  $\tilde{x} = \min_{x \in \mathbb{R}^d} \|SAx - Sb\|_2$  exactly
- Return  $\tilde{x}$
- **Observe:** if  $S$  is an OSE then

$$\|A\tilde{x} - b\|_2 \leq (1 + O(\varepsilon)) \min_{x \in \mathbb{R}^d} \|Ax - b\|_2 \quad (\text{why?})$$

- **Hence:** Our next goal is to build OSEs (ideas?)



# JL => IPJL

- **Definition (“usual” JL):** a distribution over random matrices  $S \in \mathbb{R}^{m \times n}$  satisfies the  $(\varepsilon, \delta, N)$ -JL property if for every given set of  $N$  vectors  $y_1, \dots, y_N \in \mathbb{R}^n$  we have:

$$\Pr \left[ \forall i, j: \|Sy_i - Sy_j\|_2 = (1 \pm \varepsilon) \|y_i - y_j\|_2 \right] \geq 1 - \delta$$

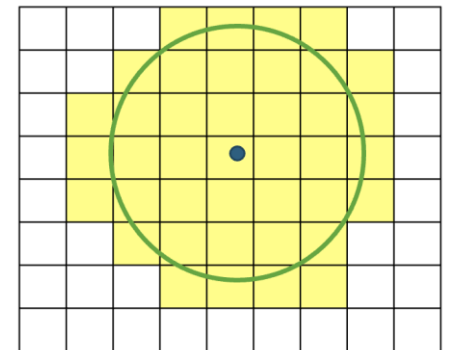
- **Definition:**  $S \in \mathbb{R}^{m \times n}$  satisfies the  $(\varepsilon, \delta, N)$ -inner product JL property (IPJL) if for every given set of  $N$  vectors  $x_1, \dots, x_N \in \mathbb{R}^n$  we have:

$$\Pr \left[ \forall i, j: |x_i^T S^T S x_j - x_i^T x_j| < \varepsilon \|x_i\|_2 \|x_j\|_2 \right] \geq 1 - \delta$$

- **Claim:**  $(\varepsilon, \delta, N^2)$ -JL implies  $(O(\varepsilon), \delta, N)$ -IPJL
  - **Proof?** The  $N^2$  vectors are every  $x_i$  and every  $x_i + x_j$  (why?)
- **Corollary:**  $S \in \mathbb{R}^{m \times n}$  with iid Gaussians and  $m = O\left(\frac{\log(N/\delta)}{\varepsilon^2}\right)$  satisfies  $(\varepsilon, \delta, N)$ -IPJL

# IPJL $\Rightarrow$ OSE

- Let  $S$  be a random matrix satisfying  $(\varepsilon, \delta, N)$ -IPJL
- Let  $\mathcal{V} \subset \mathbb{R}^n$  be a subspace of dimension  $d$
- We want to show: w.p.  $1 - \delta$ ,  $\forall x \in \mathcal{V}$ ,  $\|Sx\|_2^2 = (1 \pm \varepsilon)\|x\|_2^2$
- It suffices to show this for all unit vectors  $x \in \mathcal{V}$  (why?)
- **Definition:** an  $\eta$ -net for the unit sphere in  $\mathcal{V}$  is a subset  $Z \subset \mathcal{V}$  such that for every unit vector  $x \in \mathcal{V}$ , there is  $z \in Z$  such that  $\|x - z\|_2 \leq \eta$ .
- **Lemma:** The unit sphere in  $\mathcal{V}$  has an  $\eta$ -net of size  $|Z| \leq O\left(\frac{1}{\eta}\right)^d$ .
  - **Proof?** We've seen this!
  - We will take  $\eta = 1/2$ , so  $|Z| = O(1)^d$



# OSE => Sketch-and-Solve Regression

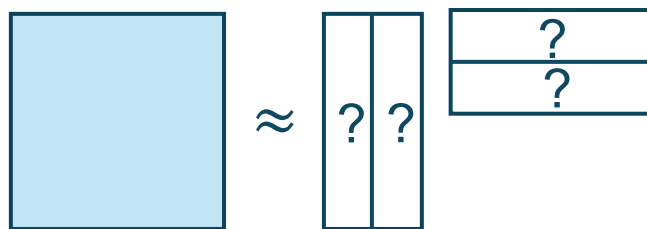
- **Corollary:**

- Let  $A \in \mathbb{R}^{n \times d}$ ,  $b \in \mathbb{R}^d$  be a regression problem.
- Let  $m = \tilde{O}(d/\varepsilon^2)$ .
- Let  $S \in \mathbb{R}^{m \times n}$  be a random matrix with iid Gaussian entries  $\frac{1}{\sqrt{m}} N(0,1)$ .
- Then, the sketch-and-solve solution  $\tilde{x} = \min_{x \in \mathbb{R}^d} \|SAx - Sb\|_2$  satisfies the approximate regression guarantee:

$$\|A\tilde{x} - b\|_2 \leq (1 + O(\varepsilon)) \min_{x \in \mathbb{R}^d} \|Ax - b\|_2$$

- What is the running time?
- See a big problem here?
- See a way around it?

# Approximate Low Rank Approximation





# Matrix Low Rank Approximation (LRA) (1/2)

- **Definition** – matrix low-rank approximation:

- Input: matrix  $A \in \mathbb{R}^{n \times d}$  of arbitrary rank, and target rank  $k \ll \min\{n, d\}$
- Goal: Find a matrix  $A' \in \mathbb{R}^{n \times d}$  of rank at most  $k$  closest to  $A$  in some metric

$$A' = \operatorname{argmin}_{\operatorname{rank}(A') \leq k} \|A - A'\|$$

- **What metric?**

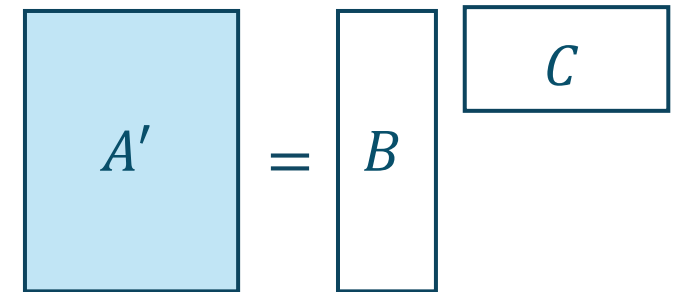
- Common choices: Frobenius norm  $\|A - A'\|_F$ , spectral norm  $\|A - A'\|_2$
- We will focus on Frobenius norm

# Matrix Low Rank Approximation (LRA) (2/2)

- Every  $A' \in \mathbb{R}^{n \times d}$  of rank  $k$  can be written as  $A' = BC$  where  $B \in \mathbb{R}^{n \times k}$  and  $C \in \mathbb{R}^{k \times d}$ 
  - How?

- Therefore, low-rank approximation is equivalent to:

$$B, C = \operatorname{argmin}_{B \in \mathbb{R}^{n \times k}, C \in \mathbb{R}^{k \times d}} \|A - BC\|$$



A diagram illustrating the matrix equation  $A' = BC$ . On the left is a light blue square box labeled  $A'$ . To its right is an equals sign. Further right is a tall, narrow white rectangular box labeled  $B$ . To the right of  $B$  is another white rectangular box labeled  $C$ , which is shorter than  $B$  and wider than  $B$ . The boxes are arranged to visually represent the multiplication of matrix  $B$  by matrix  $C$  to approximate matrix  $A'$ .

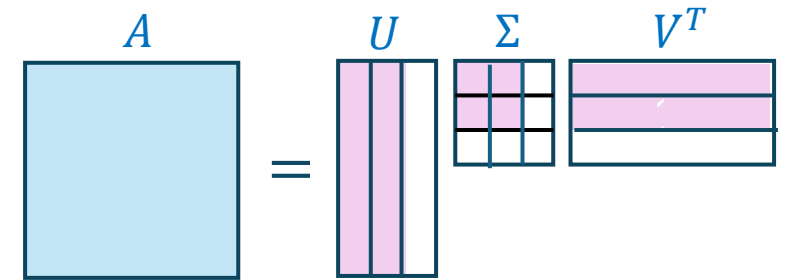
- Why LRA?
  - How much space it takes to store  $A$  vs.  $B, C$ ?
  - Given  $x \in \mathbb{R}^d$ , how much time it takes to compute  $Ax$  vs.  $BCx$ ?

# Optimal LRA via SVD (Eckhart-Young-Mirsky Theorem)

- **Recall – SVD:** Every  $A \in \mathbb{R}^{n \times d}$  of rank  $k$  can be written as  $A = U\Sigma V^T$  such that:

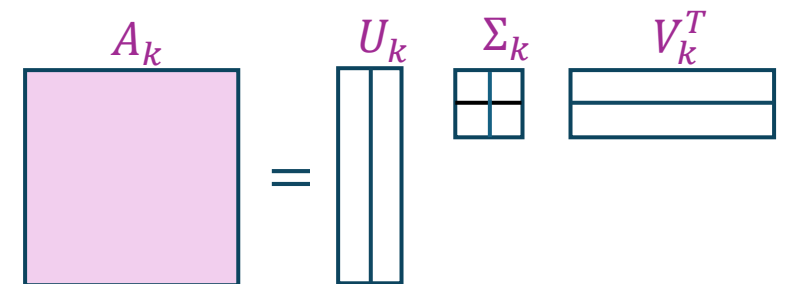
- $U \in \mathbb{R}^{n \times k}$  has orthonormal columns
- $V \in \mathbb{R}^{d \times k}$  has orthonormal columns
- $\Sigma \in \mathbb{R}^{k \times k}$  is diagonal with sorted entries :

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k > 0$$



- Equivalent writing:  $A = \sum_{i=1}^r \sigma_i \cdot u_i v_i^T$
- Let  $k < r$  be the target low rank
- Form  $A_k$  by taking the first  $r$  rows/cols of  $U, \Sigma, V$

$$A_k = \sum_{i=1}^k \sigma_i \cdot u_i v_i^T \text{ (sum is only up to } k \text{)}$$



- **Theorem:**  $A_k$  is an optimal rank- $r$  LRA for  $A$  in both Frobenius and spectral norm
  - Intuition?

# Approximate LRA

- We just saw: optimal LRA can be computed via SVD
- Consequences:
  - Optimal LRA can be computed in polynomial time (what's the running time?)
  - Optimal rank- $k$  LRA has Frobenius error  $\|A - A_k\|_F$

- **Definition – randomized approximate LRA:**

- Input: matrix  $A \in \mathbb{R}^{n \times d}$  of arbitrary rank, and target rank  $k \ll \min\{n, d\}$
- Goal: Find  $A' \in \mathbb{R}^{n \times d}$  that with probability  $1 - \delta$  satisfies:

$$\|A - A'\|_F \leq (1 + \varepsilon)\|A - A_k\|_F$$

- **Our goal**: Approx LRA algorithm in time  $O(nd \cdot (k/\varepsilon)^{O(1)})$  (linear in the size of  $A$ )

# Approximate LRA via Sketching

- Let  $A \in \mathbb{R}^{n \times d}$  and target rank  $k$  be the inputs to LRA
- Let  $S \in \mathbb{R}^{m \times n}$  be a sketching matrix, where we will choose  $k \ll m \ll n$
- What does “sketch and solve” mean in LRA? (Can we just return an LRA of  $SA$ ?)
- Let  $A' \in \mathbb{R}^{n \times d}$  have rank  $k$
- Then, the rows of  $A'$  lie in a subspace of  $\mathbb{R}^n$  of dimension  $k$  (why?)
- **Exact LRA** can choose the best rows for  $A'$  from all of  $\mathbb{R}^n$
- **Exact LRA** will choose the best rows from  $A'$  from the row-space of  $A$  (why?)
- **Approximate LRA** will choose the best rows from  $A'$  from the row-space of  $SA$ 
  - Intuition:  $\dim(SA) = m$ 
    - Since  $m \gg k$ , we have enough richness from which to choose rows for  $A'$
    - Since  $m \ll n$ , choosing the best rows for  $A'$  could be efficient

# Approximate LRA via Sketching

- **Algorithm “plan”:**

- Sample a sketching matrix  $S \in \mathbb{R}^{m \times n}$
- Compute  $SA$
- Compute and return the best rank- $k$  LRA of  $A$  within the row-space of  $SA$ :

$$A' = \operatorname{argmin}_{A': \text{rows}(A') \subset \text{span}(\text{rows}(SA))} \|A - A'\|_F$$

- **Claim:** Every  $A' \in \mathbb{R}^{n \times d}$  of rank  $k$  in the row-space of  $SA$  is equal to  $A' = BSA$  where  $B \in \mathbb{R}^{n \times m}$  is some matrix of rank  $k$
- **Hence:** we are looking for the best  $B \in \mathbb{R}^{n \times m}$  of rank  $k$ :

$$A' = BSA = \operatorname{argmin}_{B \in \mathbb{R}^{n \times m}, \text{rank}(B)=k} \|A - BSA\|_F$$

# Approximate LRA via Sketching

- **Algorithm “plan”:**

- Sample a sketching matrix  $S \in \mathbb{R}^{m \times n}$
- Compute  $SA$
- Compute and return the best rank- $k$  LRA of  $A$  within the row-space of  $SA$ :

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- **Hence:** we are looking for the best  $B \in \mathbb{R}^{n \times m}$  of rank  $k$ :

$$A' = BSA = \operatorname{argmin}_{B \in \mathbb{R}^{n \times m}, \text{rank}(B)=k} \|A - BSA\|_F$$

- **What are the two big gaps we need to fill?**

- **Prove efficiency:** Given  $SA$ , how can we find the best  $B$  efficiently?
- **Prove accuracy:** How do we show the best  $B$  satisfies approximate LRA:

$$\Pr[\|A - A'\|_F \leq (1 + \varepsilon)\|A - A_k\|_F] \geq 1 - \delta ?$$

# Best LRA in Subspace

- Let  $SA = U_S \Sigma_S V_S^T$  be the SVD of  $SA$ 
  - **Note:** This is the SVD of  $SA$ , not  $S$
- Observe: the row-space of  $SA$  is equal to the row-space of  $V_S^T$  (why?)
- Therefore: We are looking for  $C$  that minimizes  $\|A - CV_S^T\|_F$
- **Theorem**: The best  $C$  is  $C = [AV_S]_k$ .
- Hence: The best rank- $k$  LRA of  $A$  within the row-space of  $SA$  is  $A' = [AV_S]_k V_S^T$ .

- **This gives us our concrete LRA-via-sketching algorithm:**

1. Sample sketching matrix  $S \in \mathbb{R}^{m \times n}$
2. Compute the SVD  $SA = U_S \Sigma_S V_S^T$  of  $SA$
3. Compute  $[AV_S]_k$
4. Return  $A' = [AV_S]_k V_S^T$

Running  
time?



# Best LRA in Subspace: Proof

- **Theorem:** The best  $B$  is  $C = [AV_S]_k$ .
- We need to show: For all  $C$  of rank  $k$ ,  $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$ .

# Best LRA in Subspace: Proof

- **Theorem:** The best  $B$  is  $C = [AV_S]_k$ .
- We need to show: For all  $C$  of rank  $k$ ,  $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$ .
- **Proof:**

# Best LRA in Subspace: Proof

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- We need to show: For all  $C$  of rank  $k$ ,  $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$ .
- **Proof:**
  - **Claim:** For every matrix  $M$ , we have  $\|M\|_F = \|MV_S^T\|_F$ . (Proof?)

# Best LRA in Subspace: Proof

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- **Proof:**
  - **Claim:** For every matrix  $M$ , we have  $\|M\|_F = \|MV_S^T\|_F$ . (Proof?)
  - Let  $C \in \mathbb{R}^{n \times m}$  have rank  $k$ .

# Best LRA in Subspace: Proof

- **Theorem:** The best  $B$  is  $C = [AV_S]_k$ .
- We need to show: For all  $C$  of rank  $k$ ,  $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$ .
- **Proof:**
  - **Claim:** For every matrix  $M$ , we have  $\|M\|_F = \|MV_S^T\|_F$ . (Proof?)
  - Let  $C \in \mathbb{R}^{n \times m}$  have rank  $k$ .
  - We have:
$$\underbrace{\|AV_S V_S^T - [AV_S]_k V_S^T\|_F}_{\text{Claim}} = \underbrace{\|AV_S - [AV_S]_k\|_F}_{\text{Why?}} \leq \underbrace{\|AV_S - C\|_F}_{\text{Claim}} = \|AV_S V_S^T - CV_S^T\|_F$$

# Best LRA in Subspace: Proof

- **Theorem:** The best  $B$  is  $C = [AV_S]_k$ .

- We need to show: For all  $C$  of rank  $k$ ,  $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$ .

- **Proof:**

- **Claim:** For every matrix  $M$ , we have  $\|M\|_F = \|MV_S^T\|_F$ . (Proof?)

- Let  $C \in \mathbb{R}^{n \times m}$  have rank  $k$ .

- We have:

$$\underbrace{\|AV_S V_S^T - [AV_S]_k V_S^T\|_F}_{\text{Claim}} = \underbrace{\|AV_S - [AV_S]_k\|_F}_{\text{Why?}} \leq \underbrace{\|AV_S - C\|_F}_{\text{Claim}} = \|AV_S V_S^T - CV_S^T\|_F$$

- Hence:

- $\|A - [AV_S]_k V_S^T\|_F = \|(A - [AV_S]_k V_S^T) V_S V_S^T\|_F + \|(A - [AV_S]_k V_S^T)(I - V_S V_S^T)\|_F$
- $= \|AV_S V_S^T - [AV_S]_k V_S^T\|_F + \|A - AV_S V_S^T\|_F$
- $\leq \|AV_S V_S^T - CV_S^T\|_F + \|A - AV_S V_S^T\|_F$
- $= \|A - CV_S^T\|_F.$



# Best LRA in Subspace: Proof

- **Theorem:** The best  $B$  is  $C = [AV_S]_k$ .
- We need to show: For all  $C$  of rank  $k$ ,  $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$ .

- **Proof:**

- **Claim:** For every matrix  $M$ , we have  $\|M\|_F = \|MV_S^T\|_F$ . (Proof?)

- Let  $C \in \mathbb{R}^{n \times m}$  have rank  $k$ .

- We have:

$$\|AV_S V_S^T - [AV_S]_k V_S^T\|_F = \|AV_S - [AV_S]_k\|_F \leq \|AV_S - C\|_F = \|AV_S V_S^T - CV_S^T\|_F$$

Claim
Why?
Claim

- Hence:

- $\|A - [AV_S]_k V_S^T\|_F = \|(A - [AV_S]_k V_S^T)V_S V_S^T\|_F + \|(A - [AV_S]_k V_S^T)(I - V_S V_S^T)\|_F$
- $= \|AV_S V_S^T - [AV_S]_k V_S^T\|_F + \|A - AV_S V_S^T\|_F$
- $\leq \|AV_S V_S^T - CV_S^T\|_F + \|A - AV_S V_S^T\|_F$
- $= \|A - CV_S^T\|_F$ .

???



# Best LRA in Subspace: Proof

- **Theorem:** The best  $B$  is  $C = [AV_S]_k$ .

- We need to show: For all  $C$  of rank  $k$ ,  $\|A - [AV_S]_k V_S^T\|_F \leq \|A - CV_S^T\|_F$ .

- **Proof:**

- **Claim:** For every matrix  $M$ , we have  $\|M\|_F = \|MV_S^T\|_F$ . (Proof?)

- Let  $C \in \mathbb{R}^{n \times m}$  have rank  $k$ .

- We have:

$$\|AV_S V_S^T - [AV_S]_k V_S^T\|_F = \|AV_S - [AV_S]_k\|_F \leq \|AV_S - C\|_F = \|AV_S V_S^T - CV_S^T\|_F$$

Claim
Why?
Claim

- Hence:

$$\begin{aligned} \|A - [AV_S]_k V_S^T\|_F &= \|(A - [AV_S]_k V_S^T) V_S V_S^T\|_F + \|(A - [AV_S]_k V_S^T)(I - V_S V_S^T)\|_F \\ &= \|AV_S V_S^T - [AV_S]_k V_S^T\|_F + \|A - AV_S V_S^T\|_F \\ &\leq \|AV_S V_S^T - CV_S^T\|_F + \|A - AV_S V_S^T\|_F \\ &= \|A - CV_S^T\|_F. \end{aligned}$$

Pythagorean  
identity





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- Next, we need to show **accuracy**: For a good choice of random sketching matrix  $S$ ,
$$\Pr[\|A - [AV]_k V^T\|_F \leq (1 + \varepsilon) \|A - A_k\|_F] \geq 1 - \delta.$$

# LRA via Sketching: Accuracy



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- Recall we saw  $(\varepsilon, \delta)$ -AMM with  $m \approx 1/\varepsilon^2$  rows (ignoring  $\delta$ )
- Corollary:**  $(\varepsilon, \delta)$ -approximate LRA in time  $O(nd \cdot (k/\varepsilon)^{O(1)})$ .
  - Linear in the size of  $A$

# LRA Accuracy: Proof

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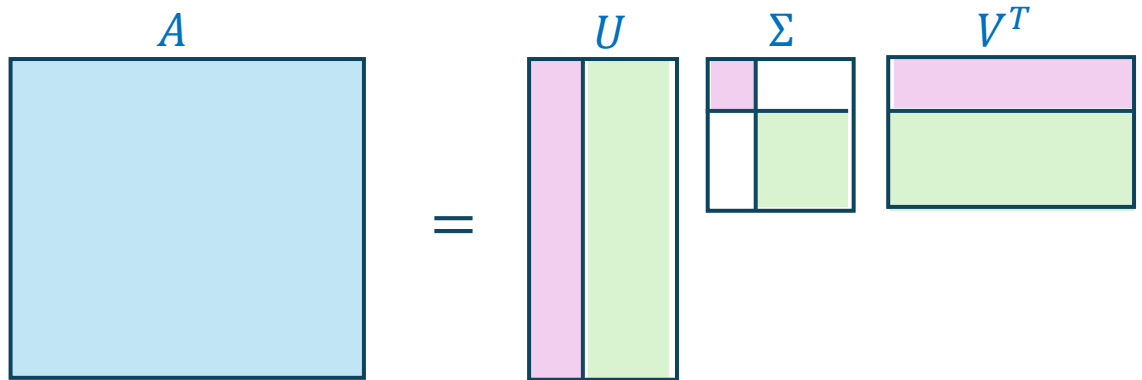
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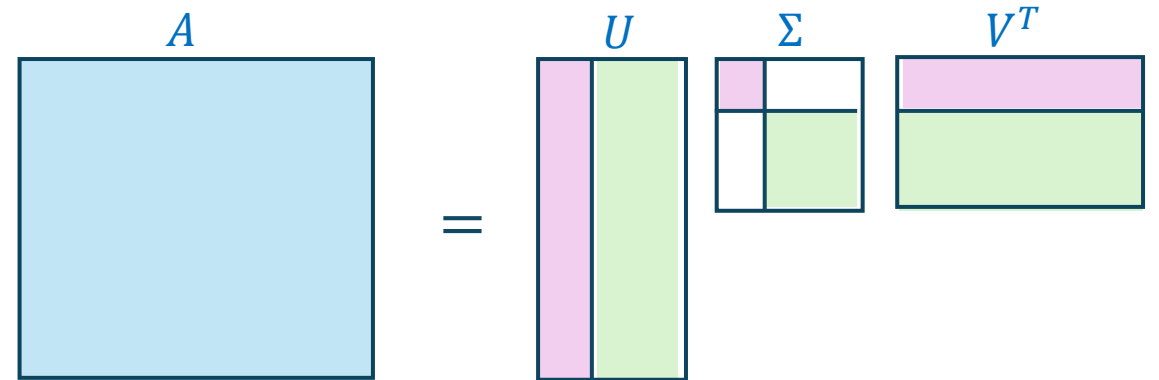
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**Observe:**

$$A_k = U_k \Sigma_k V_k^T$$
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- Note: we have assumed  $Z$  is invertible; we will soon prove this is true

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  - $= k \cdot \|Z^{-1}\|_2^2 \cdot \|U_k^T S^T S (A - A_k)\|_F^2$
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- To finish the proof, we will bound both  $\|Z^{-1}\|_2^2$  and  $\|U_k^T S^T S (A - A_k)\|_F^2$  using AMM

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- We choose  $\varepsilon$  small enough so that  $\|U_k^T S^T S U_k - I\|_2^2 \leq 1/9$



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- **Corollary:** All eigenvalues of  $Z$  are between  $2/3$  and  $4/3$ .
- In particular:
  - $Z$  is invertible.
  - $\|Z^{-1}\|_2^2 \leq \frac{16}{9} = O(1)$ .

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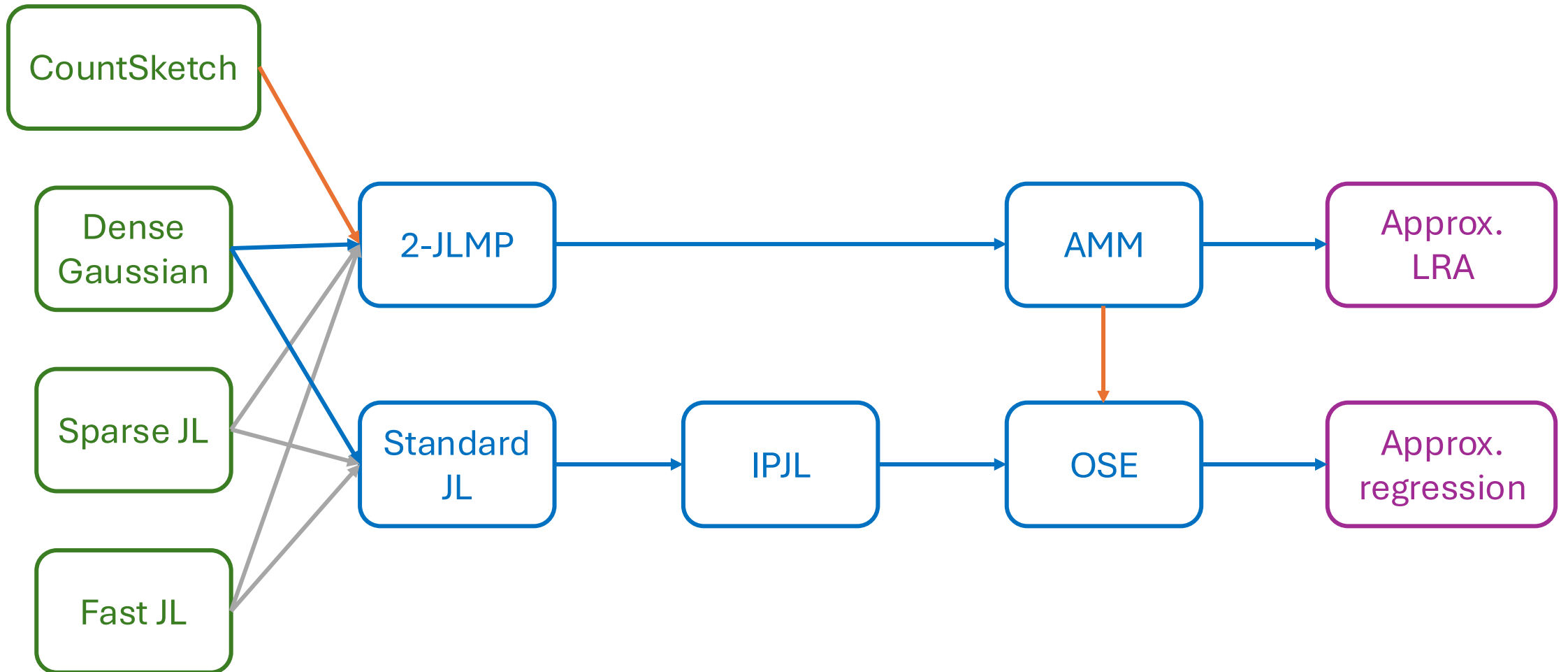


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- Which together yields:  $\|U_k Z^{-1} U_k S^T (A - A_k)\|_F^2 \leq O(k^2\varepsilon)\|A - A_k\|_F^2$  as needed.



# Efficient Linear Algebra: Summary



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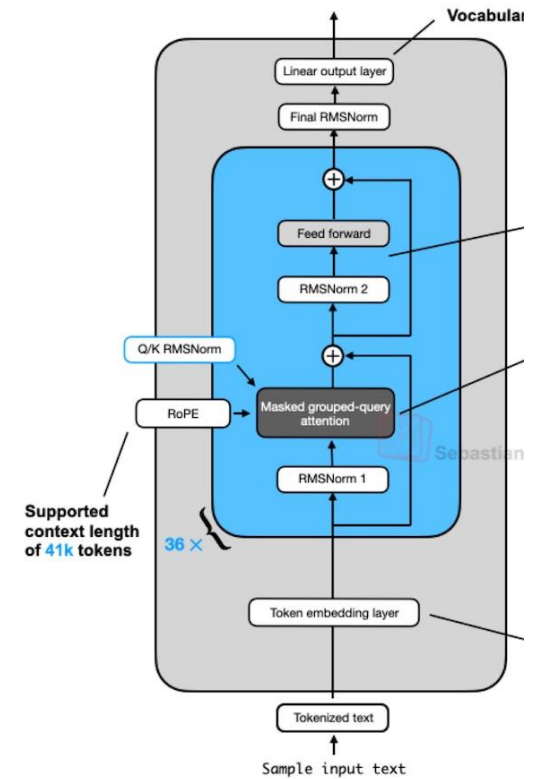
- JL and JL-like techniques are useful for randomized approximate algorithms for linear algebra problems
  - Matrix multiplication
  - Regression
  - Low rank approximation

# LoRA: Low Rank Adaptation in LLMs

An Aside

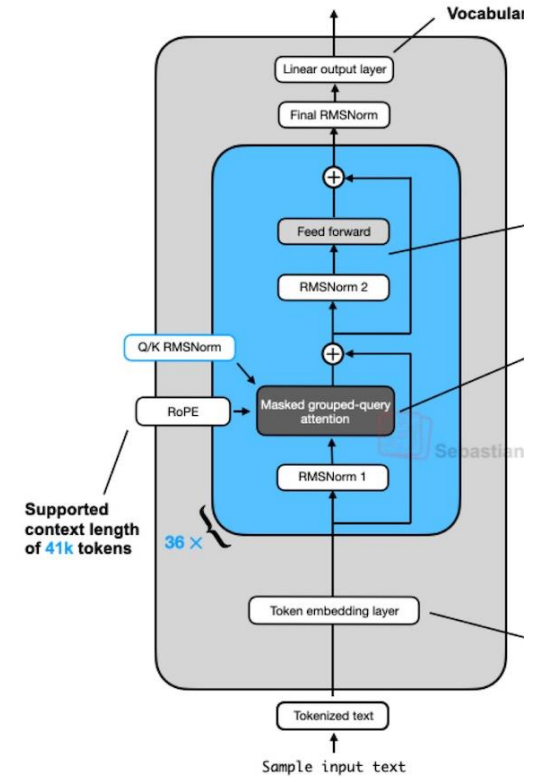
# Recap: Attention in LLMs

- Attention:
  - Input:  $K, Q, V \in \mathbb{R}^{n \times d}$
  - Output:  $\text{softmax}\left(\frac{1}{\sqrt{d}} QK^T\right) V$



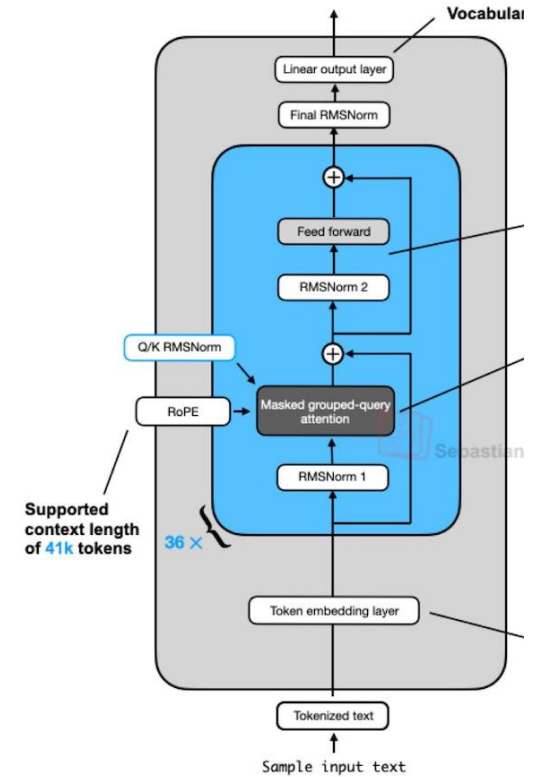
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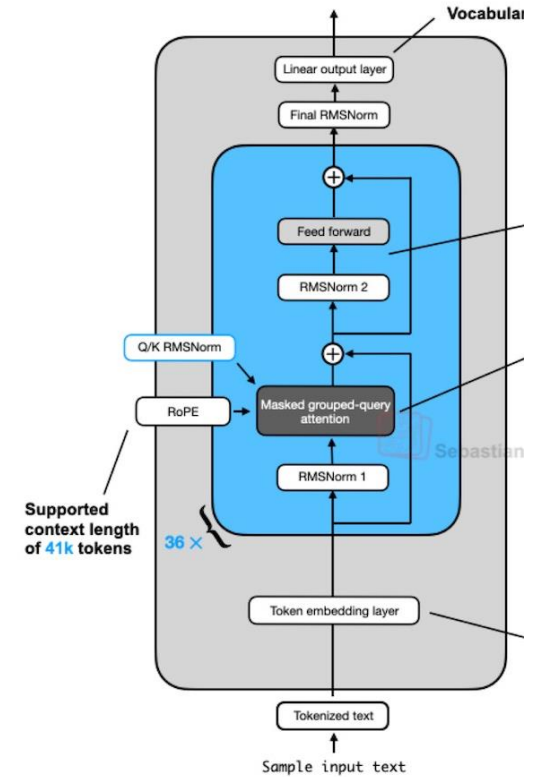
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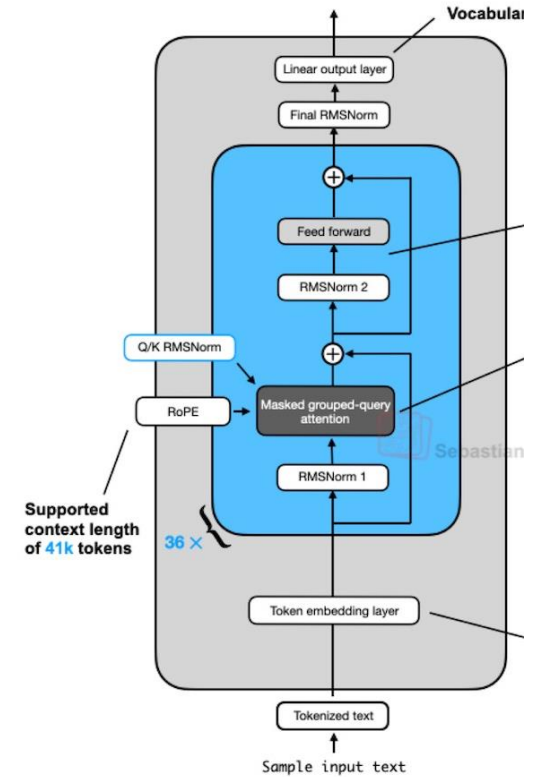
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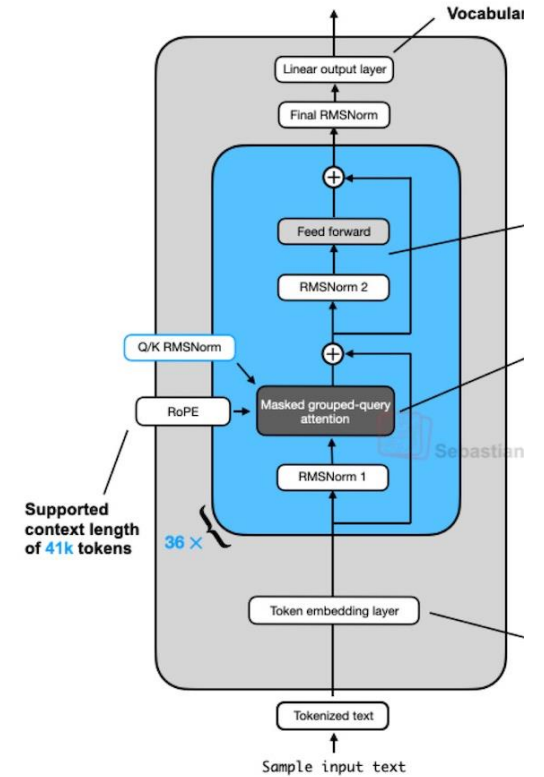
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  - Upon getting  $X$  it computes:  $K = XW_K, Q = XW_Q, V = XW_V$



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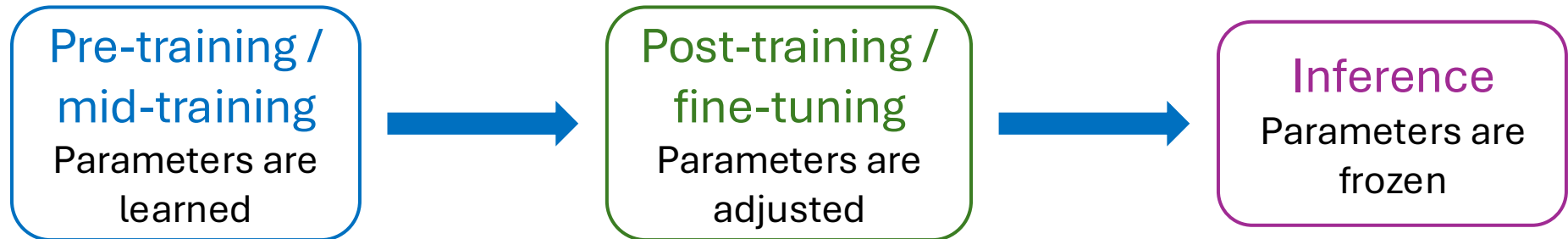


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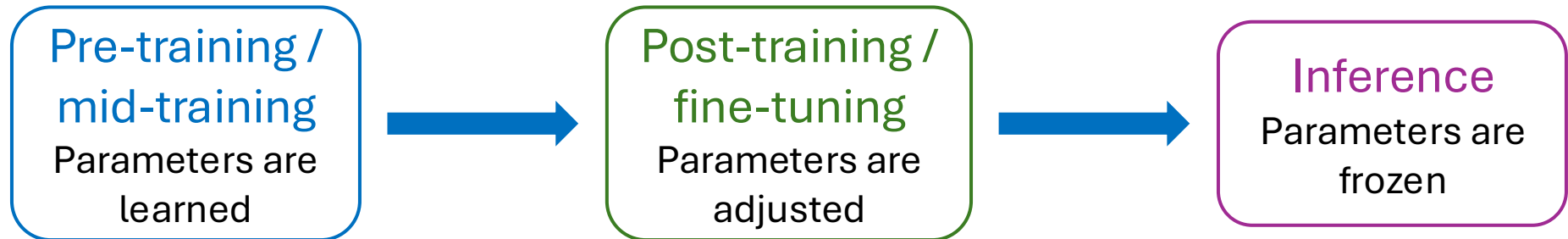


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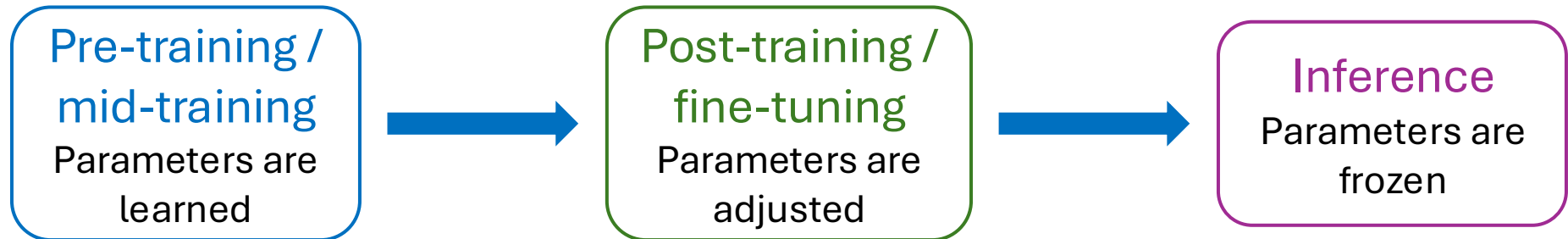
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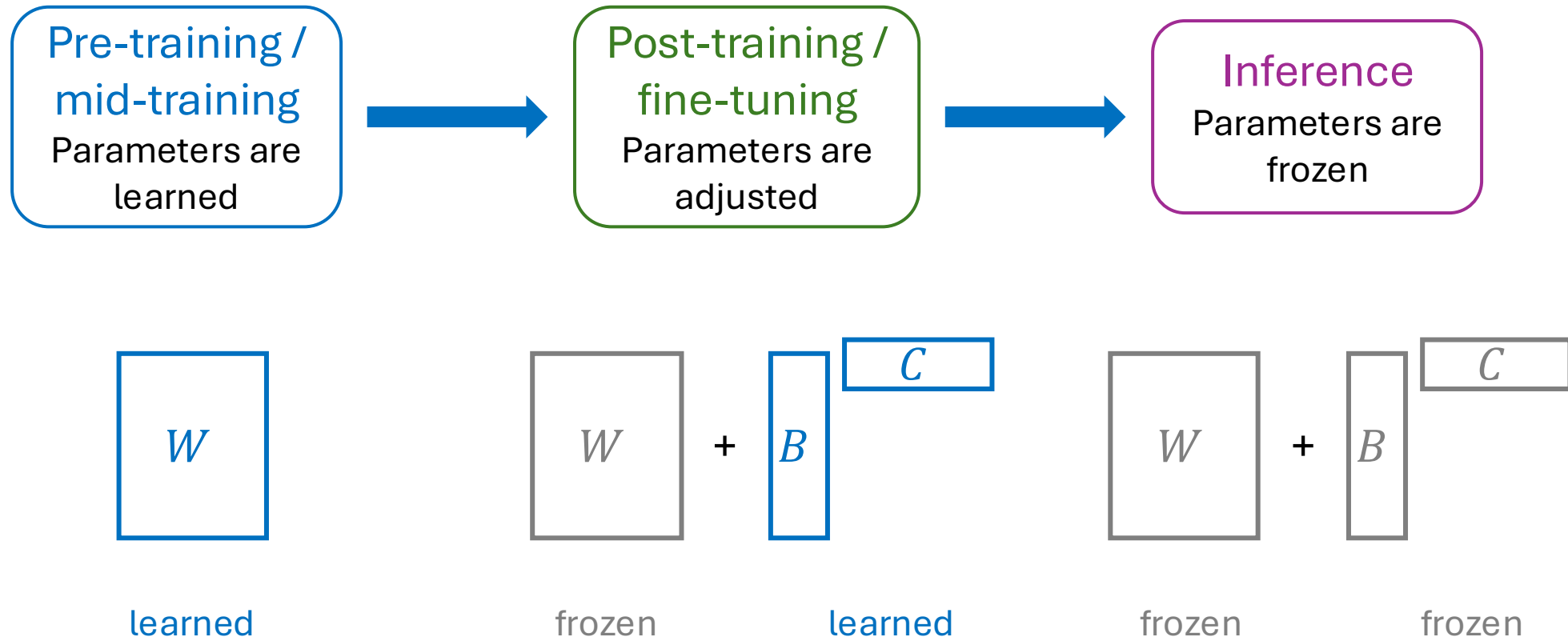


- LLMs:



- During post-training / fine-tuning, we want to adjust a small number of parameters
  - Why?

# LoRA: Low Rank Adaptation





Questions?